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A Mild Protecting-Group Free Strategy for Neoglycoconjugate Synthesis

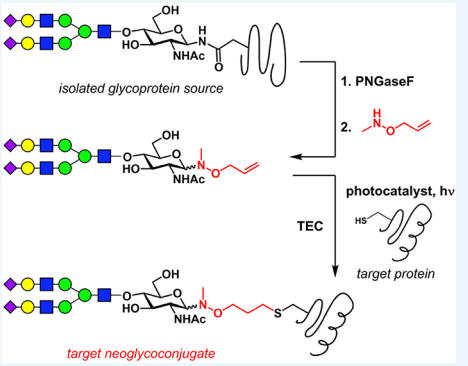
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ABSTRACT: The synthesis of neoglycoconjugates has paved the way for the discovery of novel probes that mimic natural glycoconjugates and can provide designed research tools and therapeutics. In some cases, the target protein may not be amenable to harsh conditions; therefore, semisynthetic or chemical methods must be chosen with care. Here, we present a simple and modular chemoselective coupling strategy between an unprotected sugar and an *N*,*O*-disubstituted hydroxylamine under mild acidic conditions. This strategy removes any need for protecting groups on the glycan. The terminal alkene group of the conjugate serves as an effective handle to allow facile conjugation to the protein of interest via thiol−ene coupling (TEC), with proteins bearing a cysteine or free thiol to prepare neoglycoconjugates. We demonstrate that the strategy is compatible with both N- and O-linked glycans using protecting-group free strategies and optimize the TEC conditions using a variety of photocatalysts. Finally, we test the method on an aggregation-prone protein, α-synuclein. We envision that this strategy could allow the construction of complex glycoconjugates for biological testing using isolated glycans, or for generation of conjugates where the protein of interest is sensitive to harsh conditions.



INTRODUCTION

Glycoproteins are critical components of the cell, and their glycans may help maintain structure, enhance solubility, or provide recognition.[1] One of the challenges in understanding the biological role of glycoproteins is the heterogeneity of the conjugates, as multiple glycan structures may be appended to the same protein (i.e., proteoforms).[2] A primary tool to investigate glycoconjugate function is the design of synthetic analogs, known as neoglycoconjugates (NGCs). Strategies that provide well-defined glycan structures on a protein of interest can be used to investigate the biological role of glycan structure in assays and disease models.[3] Asparagine-linked glycans are nearly ubiquitous modifications found on glycoproteins, and synthesis of NGCs that mimic this common attachment are important targets.

Synthetic strategies for the construction of a well-defined NGC include a range of chemical and chemoenzymatic approaches.[4−6] These methods often require the use of harsh reaction conditions and protecting groups on intermediates. Cell-based methods may engineer the glycosylation pathway using gene editing tools[7] or expression vector libraries,[8] and serve as useful resources for chemoenzymatic synthesis of oligosaccharides and glycoproteins. Although diverse proteoforms can be generated using different glycosyltransferases, it is often a challenge to obtain homogeneous glycoproteins.[9] Various mild ligation strategies have been used to attach glycans to NGCs including aminooxy,[10] hydrazide,[11] native chemical ligation (NCL),[12] 'tag and modify',[13] and expressed protein glycoligation (EPG).[14] Strategies that exploit reductive amination are commonly used for generating NGCs, but result in loss of the closed-ring form of the reducing end residue which may impede or abolish recognition by lectins.[15,16] The use of *N*-alkoxyamines can provide access to neoglycosides with a closed ring at the reducing end.[17−19] Chemoenzymatic methods have been developed to assemble NGC, but are subject to substrate specificity of enzymes and often require synthetic manipulations of the glycan.[20] Auxiliary-based ligation has enabled the construction of large natural glycoproteins; however, some target glycoproteins may be challenging and choice of an optimal ligation strategy with minimal manipulations is essential.[21]

Many glycoprotein targets of interest may be difficult to work with due to instability or sensitivity to harsh chemical conditions. Several approaches have been developed to generate defined NGCs that require minimal protein or glycan manipulations. For example, proteins or peptides featuring an amino acid residue that includes an *N*-methylaminooxy residue can be used to directly conjugate unprotected glycans.[22] A UV-

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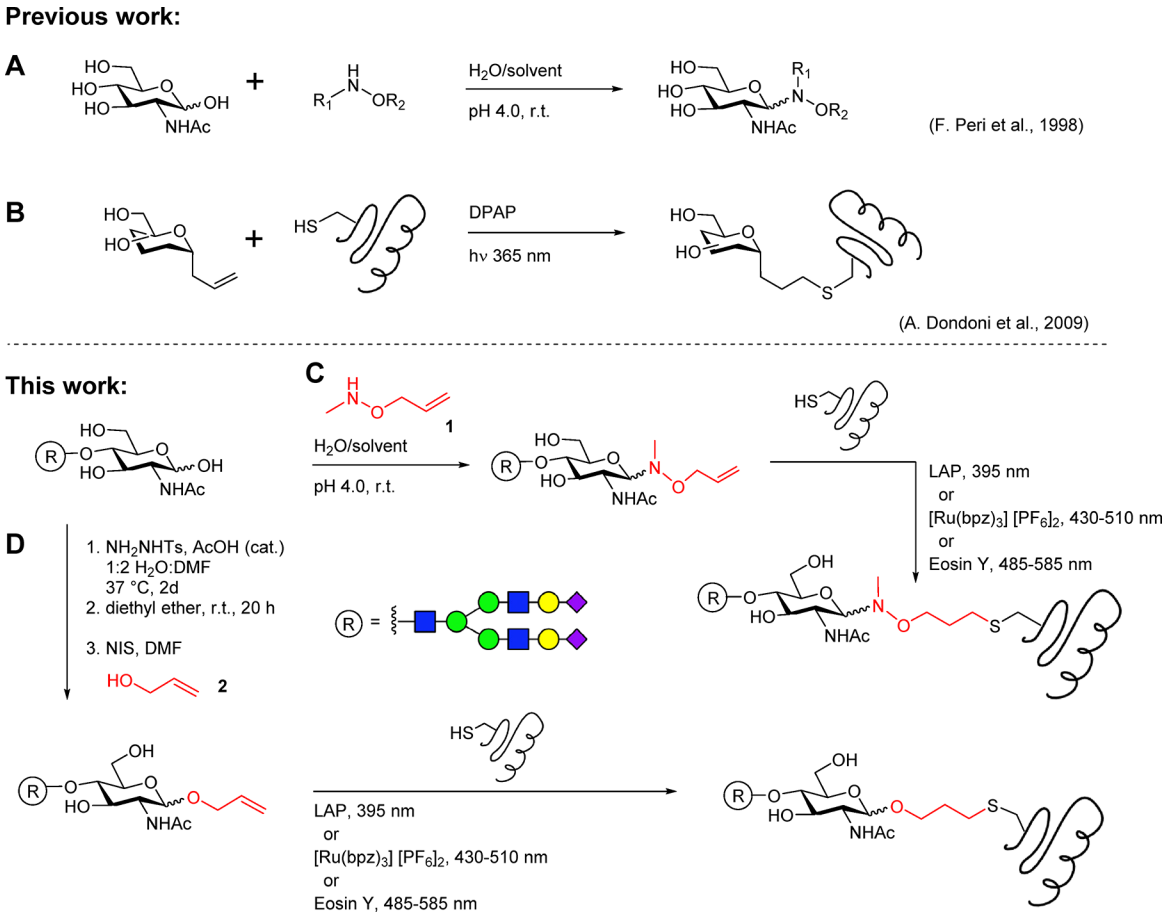
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Scheme 1. Previous Protecting-Group Free Approaches to (A) *N*-Glycosidation[19] and (B) TEC to Proteins;[29] Here We Develop (C) an *N,O*-Dialkylated Hydroxylamine Bifunctional Linker, and (D) a Two-Step *O*-Glycosidation Strategy[31]



activated urea-based linker has been employed to directly conjugate glycosylamines.[23] Synthetic peptides or proteins with an installed thioester function (ThioD) can undergo NCL reactions with glycans featuring an auxiliary.[24] Due to the low natural abundance of cysteine in proteins, coupling strategies that exploit cysteine can result in more homogeneous neoglycoproteins with fewer manipulations. Site-directed mutagenesis of cysteine on a protein of interest can provide a specific handle for conjugation.[25] Synthetic neoglycopolymers bearing multiple copies of sugar moieties have been conjugated to thiol-bearing proteins using a maleimide linker.[26] Thiol−ene coupling (TEC) is a click-reaction that relies on photoinduced radical addition of thiols to alkenes.[27−29]

We considered that the combination of TEC strategies with protecting-group free glycosidation strategies could be used to produce NGCs with sensitive protein targets that would be valuable for biological testing or development of therapeutics. In this work, we have developed a facile, modular, and protecting-group free linker strategy to couple reducing glycans to a protein of interest using TEC (Scheme 1). At present, there are few examples that combine protecting-group free glycosidation with *N*-alkylhydroxylamines and click bioconjugate additions like TEC.[3,18,23,30] Central to the strategy used are readily available linkers: *O*-allyl-*N*-methylhydroxylamine **1** or allyl alcohol **2**. These linkers can be installed under mild conditions on reducing sugars which may be synthetic or isolated from a biological source. The allyl group of the linkers can then be used as an effective handle to allow facile conjugation to a protein of interest via TEC. This strategy can be applied to synthetic or purified glycans to generate defined NGCs. To mimic linkages of N-linked glycans, we performed model conjugations with GlcNAc, *N,N′*-diacetylchitobiose, and a biantennary N-link glycan. We optimized the TEC reaction using different UV- and visible light-based radical initiators and confirmed the feasibility of conjugation to different protein targets. Multiple initiators encompassing a range of light frequencies for activation were tested as those that can be activated at longer wavelengths are less likely to cause protein degradation.[29] To illustrate the utility of the strategy, we applied our conjugation strategy to the modification of an unstable protein target, *α*-synuclein, where harsher or more complex synthetic methods may be problematic.

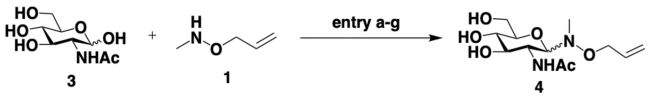
RESULTS AND DISCUSSION

Protecting-Group Free Linker Conjugation to Reducing Sugars. Previous reports have used *N,O*-dialkyl-hydroxylamines to efficiently generate glycosides, preserving the cyclic structure of the residue.[19] To provide a handle for thiol−ene coupling we identified a commercially available linker, *O*-allyl-*N*-methylhydroxylamine **1**, containing both a terminal alkene and an *N*-methyl-hydroxylamine (Scheme 1c). Compound **1** reacted with a reducing *N*-acetyl-D-glucosamine (GlcNAc) to form *O*-allyl-*N*-methyl-*N*-(2-aminoacetyl-2-deoxy-D-glucopyranosyl)hydroxylamine **4** following reported procedures[19,32−36] (Table 1). While this reaction has been reported

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Table 1. Optimization of Reaction Conditions for *N*-Glycosidation Using *N*-Acetyl-D-glucosamine 3 and *O*-Allyl-*N*-methylhydroxylamine 1



3 1 4

entry	solvent	temp., time	yield, α:β
a	1:1 (v/v) DMF/AcOH	rt, 20 h	91%, 76:24
b	5:1 (v/v) MeOH/AcOH	rt, 72 h	75%, 71:29
c	2 M aq. NH_4OAc pH 4.5 buffer	40 °C, 72 h	76%, 59:41
d	1:1 (v/v) 0.1 M NaOAc/AcOH pH 4.0 buffer/H_2O	0 °C to rt, 20 h	73%, 66:34
e	2:1:0.04 (v/v/v) CH_3CN/H_2O/AcOH	rt, 16 h	87%, 58:42
f	3:1 (v/v) DMF/AcOH	40 °C, 72 h	83%, 59:41
g	1:1 (v/v) 1,4-dioxane/NH_4OAc pH 4.5 buffer	40 °C, 72 h	95%, 51:49

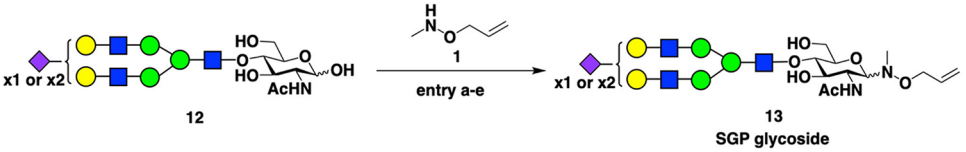
to be β-selective in most cases, anomeric mixtures have been observed.[37−40] Goff and Thorson have observed that neoglycosides of some natural products and small molecules can give variable anomeric product ratios.[40] The specific α to β anomeric ratio will depend on factors including the nature of the *N*,*O*-disubstituted hydroxylamine, the reaction conditions, and the sugar itself.[41,42] We observed that the α-product was more prevalent or gave close to 1:1 α:β mixtures in reaction of **1** with GlcNAc (yield 73−95%; Tables 1 and S1). Similar conditions with the disaccharide, *N*,*N*′-diacetylchitobiose (Scheme S2 and Table S2) gave good conversion to the glycoside, which should be representative for coupling of more complex N-link glycan structures. Although our primary focus was on the synthesis of N-link glycan mimics, we also considered that O-link glycans could be of interest. Therefore, we tested a protecting-group-free linker strategy for *O*-linked glycosides using *p*-toluenesulfonylhydrazide (Scheme 1d). We employed a simple two-step glycosidation reaction that yielded the β-pyranoside as expected after reaction with allyl alcohol **2** (1:6 α:β selectivity; Scheme S1).[31,36] Both methods were able to provide up to 90% conversion to an alkene-labeled glycoside, although we favored the *N*,*O*-alkyl-hydroxylamine strategy as it requires fewer manipulations of the glycan. We note that the *O*-glycosidation provided better anomeric selectivity and could be advantageous for targets where the reducing end residue is involved in recognition.

To demonstrate conjugation with a complex glycan, we isolated sialoglycopeptide (SGP) from egg yolk following reported protocols.[43,44] The glycan was then released from the peptide using recombinant PNGase-F and purified using solid-phase extraction. Glycosidation of the SGP glycan **12** was optimized with linker **1** to generate glycoside **13** (Table 2). Similarly, *O*-glycosidation of glycan **12** using *p*-toluenesulfonylhydrazide activation and allyl alcohol **2** provided glycoside **16** (Scheme S3). Based on these data, we favor the use of the conditions shown in entry **b** for glycosylations based on the anomeric ratio and yield (Table 2).

Thiol−Ene Coupling with Photocatalysts. Thiol−ene coupling (TEC) has been used extensively for the synthesis of defined protein conjugates,[28] and we first optimized conditions for TEC with model glycosides bearing an allyl group. Using the *O*-allyl-*N*-methylhydroxylamine **1** and allyl alcohol **2** linkers, reaction with L-cysteine (L-Cys) was successful resulting in 65 and 76% crude yields, respectively (Scheme S4). Reported TEC conditions with UV−B photoreactors (305−310 nm, 0.5−3 h) under argon atmosphere with excess cysteine (>0.5 M) are not feasible for use with proteins due to photodegradation.[45] Instead, photoinitiators should be used with longer wavelength light for protein conjugation. Testing of radical initiators commonly used for TEC, such as 2,2-dimethoxy-2-phenylacetophenone (DPAP or DMPA or Irgacure 651) and 2-hydroxy-4′-(2-hydroxyethoxy)-2-methylpropiophenone (HHMP or Irgacure 2959) were positive in the case of L-Cys coupling but did not progress using proteins and glycans (data not shown).[46] We proceeded to test initiators that absorbed at >350 nm including lithium phenyl-2,4,6-trimethylbenzoylphosphinate (LAP; λ_{max} = 380 nm),[47] tris-(2,2′-bipyrazine)ruthenium bis(hexafluorophosphate) ([Ru-(bpz)$_3$][PF$_6$]$_2$; λ_{max} = 453 nm),[48] and Eosin Y disodium salt (EY; λ_{max} = 524 nm)[49] for protein conjugation reactions (Table S3).

TEC between Lysozyme and BSA with mPEG-Alkene (10 K). To identify optimal conditions for TEC conjugation on proteins, we employed an alkene terminated PEG polymer as a 10 kDa MW (mPEG-Alkene 10 K) mass tag. With this alkene we tested TEC on model proteins, lysozyme (Lyz; containing 8 Cys residues) and bovine serum albumin (BSA; containing 35 Cys residues). We first compared conjugation conditions with and without radical initiators or irradiation using the three initiators identified above (Table S3 and Figure 1). The stability of the proteins Lyz and BSA to UV irradiation (395

Table 2. Optimization of Reaction Conditions for *N*-Glycosidation of a Complex Glycan from Sialoglycoprotein (SGP)



12 13 SGP glycoside

entry	solvent	temp., time	SGP glycoside 13 crude yield, α:β
a	1:1 (v/v) DMF/AcOH	rt, 20 h	67%, 51:49
b	5:1 (v/v) MeOH/AcOH	rt, 72 h	76%, 42:58
c	2 M aq. NH_4OAc pH 4.5 buffer	40 °C, 72 h	73%, 36:64[a]
d	1:1 (v/v) 0.1 M NaOAc/AcOH pH 4.0 buffer/H_2O	0 °C to rt, 20 h	85%, 74:26
e	2:1:0.04 (v/v/v) CH_3CN/H_2O/AcOH	rt, 16 h	65%, 28:72[a]

[a]The H1-β signal integration was affected by overlap with solvent, and the ratio may not be accurate.

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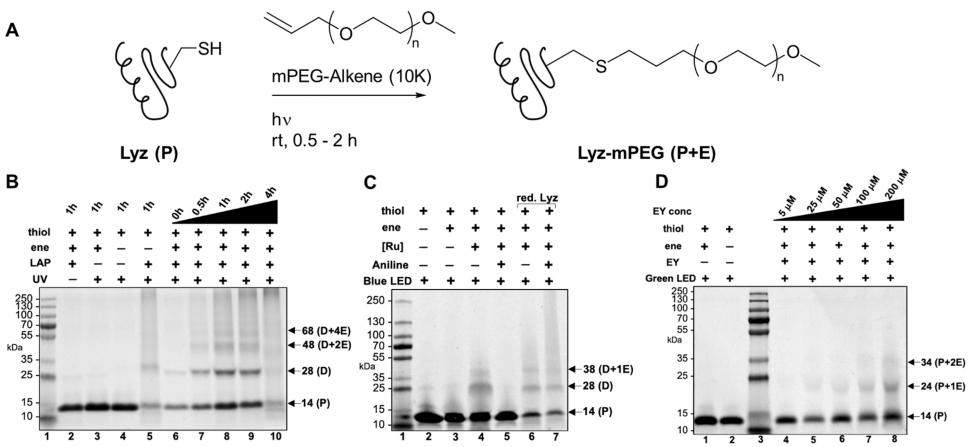


Figure 1. (A) Schematic representation of TEC conjugates with Lyz and mPEG-Alkene-10K. Initiators for TEC were EY, LAP, and [Ru]. (B) SDS-PAGE of reaction mixtures containing Lyz, mPEG-Alkene, and LAP (1.5 mM) are shown. (C) SDS-PAGE of reaction mixtures containing Lyz, mPEG-Alkene, and [Ru] (0.25 equiv) and aniline (0.5 equiv). Lyz was pretreated with TCEP prior to irradiation in lns 6 and 7. (D) SDS-PAGE of reaction mixtures containing Lyz, mPEG-Alkene, and EY (5−200 mM). Samples were loaded on SDS-PAGE (12%) under reducing conditions and visualized by Coomassie staining. Observed bands are labeled as P, Lyz protein; D, Lyz dimer; D+2E, dimer with 2 alkene units; and D+4E, dimer with 4 alkene units.

nm) was confirmed using MALDI-MS (Figure S1). Although we did not perform circular dichroism (CD) measurements, previous studies of BSA irradiated with a UV source suggests the protein maintains its gross conformation.[50] Furthermore, non-irradiated controls were included in SDS-PAGE analysis of conjugation reactions. Lyz is known to spontaneously form dimers,[51] and we observed the monomer and dimer of Lyz (14 kDa, P; 28 kDa, D) as conjugation products in both LAP and [Ru] trials (Figure 1B,C). Up to 4 mPEG units were conjugated to Lyz using LAP (1.5 mM, 1 h; Figure 1B, Ln 8). Using EY as the initiator, only the singly conjugated Lyz was observed at 24 kDa (P+1E) at 20 min of irradiation; and longer irradiation showed formation of a doubly modified Lyz (P+2E; Figure 1D, Ln 7 and 8). The appearance of new conjugate bands was observed with and without prior reduction of the protein with tris(2-carboxyethyl)phosphine (TCEP) (Figure 1C). Furthermore, higher molecular weight streaking indicated coupling to multiple thiols on Lyz in several conditions.

As Lyz may itself be glycosylated,[52] we next tested mPEG-alkene conjugation of BSA using the same conditions (Figure 2). We observed that the LAP (1.5−2 mM, 2 h) and EY (0.1−0.2 mM, 0.3 h) initiators gave apparent consumption of the protein (P, 66 kDa) and the formation of conjugates with up to 10 PEG chains attached per protein (Figure 2B, P+10E, 166 kDa). Analysis of starting material consumption by densitometry (Figures S2 and S3) confirmed that TEC with BSA and the LAP initiator gave maximum consumption of starting material (99% in lane 5, 60% in lane 7 and 90% in lane 10; Figure S3). The [Ru] initiator required longer irradiation to give significant conversion (0.17 mM, 0.7 h; Figure S3). Since the conjugation products show streaking by SDS-PAGE, we confirmed that all three initiators gave 70−90% conjugate formation using HPLC. Reactions with EY, [Ru], and LAP photocatalysts also showed the appearance of more polar conjugates (P+E, Figure 2C) relative to P. Finally, to rule out nonspecific labeling of the protein, we performed TEC conjugations on BSA pretreated with iodoacetamide (IAM; Figure S4 and Table S4). As expected, IAM pretreatment prevented formation of the alkene conjugates (P+E). With optimized conditions for TEC conjugation to proteins, we proceeded to test reactions of the alkene-labeled glycosides.

Generating TEC Neoglycoconjugates with SGP-Glycan. To generate BSA neoglycoconjugates we used the isolated SGP glycoside **13**. We tested reaction conditions using radical initiators, followed by lectin blot to detect conjugation. Conjugates were probed with biotinylated lectins peanut agglutinin (PNA) and *Erythrina cristagalli*agglutinin (ECA), which should recognize terminal LacNAc, and *Sambucus nigra* agglutinin (SNA), which recognizes sialic acid residues (Figure 3).[53] Unfortunately, BSA background was high with biotinylated SNA as a probe (data not shown) and these data were inconclusive. PNA staining was absent in unreacted BSA (Figure 3D), and TEC with **13** resulted in staining for conjugates after LAP and [Ru] catalyzed reactions (Figure 3). ECA staining was inefficient, but we observed higher MW streaking only for reactions with photoinitiators (Figures 3C and S5, Lns 8 to 10). Lectin blot analysis showed that the glycosylated forms of BSA migrated more slowly through the gel than the nonglycosylated BSA. Analysis of the conjugates by RP-HPLC detected the formation of new peaks (P+E) with faster migration and consumption of the protein for all three initiators (Figure S6). We confirmed conjugate yields (P+E) using HPLC: 69% with EY, 73% with LAP and 60% with [Ru] (Table S5). Negative controls with IAM-capped BSA or reactions without irradiation did not show any new conjugate peaks.

Neoglycoconjugate samples were also analyzed by MALDI to confirm attachment of the SGP glycan to BSA (Figure S7 and Table S6). We observed a shift in the mass of the protein by MALDI after reaction with SGP-glycoside **13**, corresponding to as many as two units of SGP linked to BSA. Terminal *N*-acetylneuraminic acids were not observed, likely due to lability of these residues in mild acidic conditions. The observed mass shift was consistent with the conjugates bearing terminal galactose residues, as detected by PNA lectin staining (Figure 3).

Synthesis of α-Synuclein Glycoconjugates. We next turned our attention to a challenging protein target that could benefit from the mild conjugation strategy developed here. The α-synuclein protein is a disordered 140-amino acid residue

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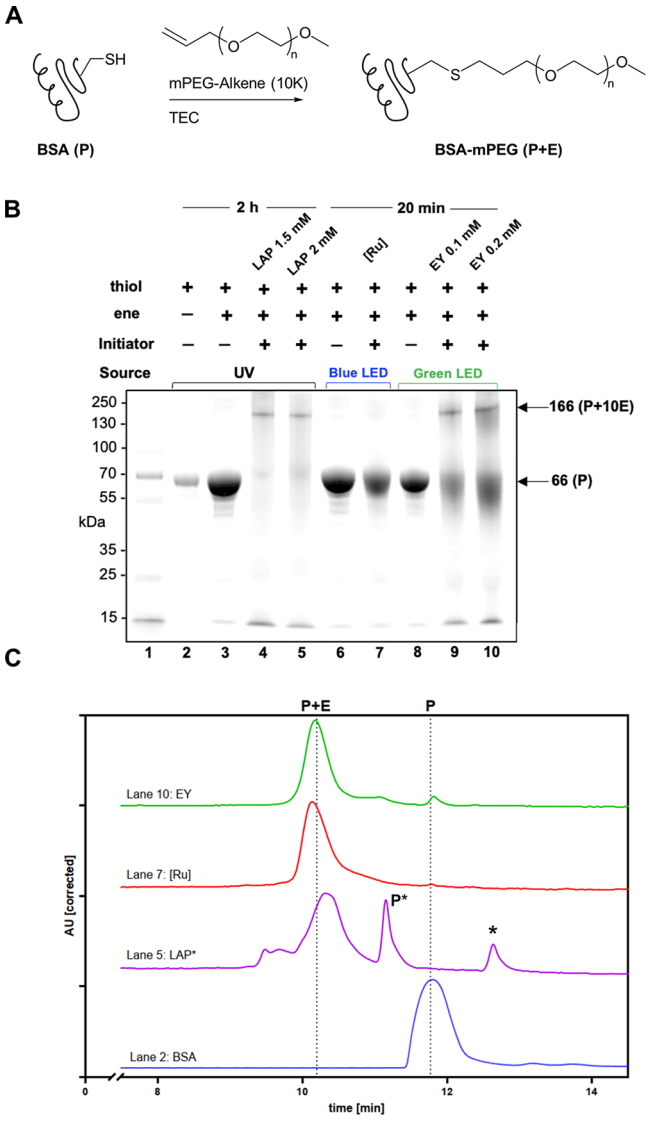


Figure 2. (A) Schematic representation for the TEC of mPEG with BSA. (B) SDS-PAGE analysis of BSA conjugates with Coomassie staining. Reactions used the radical initiators (LAP, [Ru], or EY) and a light source as indicated. Expected MW for BSA (P) and the mPEG-alkene-10K (E) are indicated. (C) HPLC analysis of TEC conjugates. Samples were from the same reactions shown in panel (B). Samples were loaded on an SDS-PAGE (10%) under reducing conditions and visualized by Coomassie staining. Observed signals are labeled as P, BSA protein; P+E, BSA-alkene conjugate; and P+10E, protein with 10 alkene units. LAP treatment shifted the protein peak to a shorter retention time as observed in lane 5, indicated as P*, and the peak corresponding to the LAP photocatalyst in the sample is indicated as '*'. The proportions of desired conjugates (P+E) to protein (P) are as follows: Ln 5—73% P+E, 15% P; Ln 7—94% P+E, 1% P; Ln 10—90% P+E, 4% P (see Table S4).

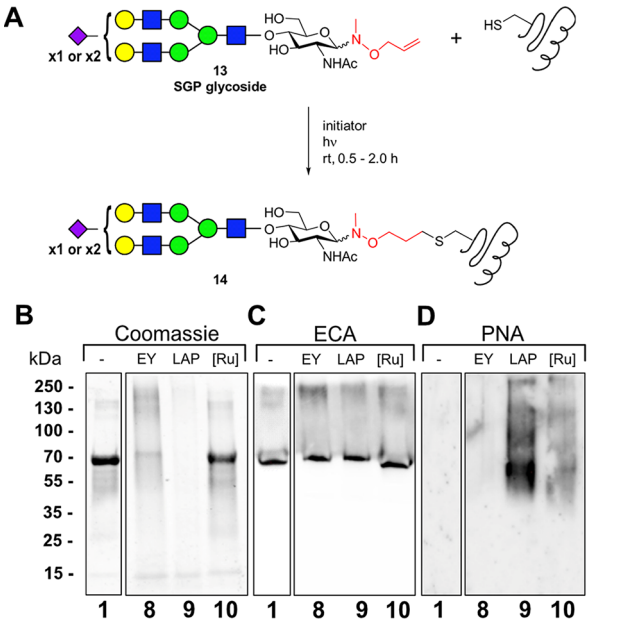


Figure 3. (A) Schematic representation for TEC between BSA and SGP-glycoside **13**. (B) SDS-PAGE with Coomassie staining. (C) Lectin blot using biotinylated-ECA. (D) Lectin blot using biotinylated-PNA. Lectin blots were developed with enhanced chemiluminescence reagent. Lane 1 represents unlabeled BSA, and lanes 8, 9, and 10 represent conjugation using EY (20 min); LAP (2 h); and [Ru] (20 min) as initiators. Observed peaks for BSA are labeled as P (MW of 66 kDa). Samples were loaded on SDS-PAGE (10%) under reducing conditions, transferred to nitrocellulose, and probed with ECA or PNA. The blot was developed with streptavidin-HRP. The full gel image is provided in Figure S5.

protein (15 kDa) known to aggregate into Lewy bodies found in Parkinson's disease brains.[54−57] Interestingly, glycation and glycosylation of α-synuclein have been suggested to influence the protein's aggregation and its biological effects.[57−59] Natively unfolded α-synuclein adopts a β-strand-rich conformation that can fold into different tertiary structures.[56] Aggregation of α-synuclein is thought to involve multiple steps, starting with nucleation of small oligomers followed by growth that leads eventually to large fibrillar aggregates.[59] The α-synuclein protein presents a challenging target for chemical manipulation as it is conformationally sensitive[60,61] and chemically unstable;[62] additionally the protein is known to aggregate or become cross-linked in the presence Cu^{II} ions commonly used for CuAAC conjugations.[63,64] There are no native cysteines in α-synuclein, thus a recombinant protein with a cysteine residue appended to the C-terminus was expressed and purified.[65] Initial trials of TEC with α-synuclein using mPEG-Alkene (10 K) showed significant consumption of the α-synuclein band by SDS-PAGE (data not shown).

We subjected a sample of α-synuclein to TEC conditions with SGP glycoside **13** and followed the conjugation by SDS-PAGE and lectin blot (Figure 4). The SDS-PAGE of the reactions showed either consumption of the unmodified protein or shifts in the apparent MW. Apparent high molecular weight streaking was increased in all conjugate conditions.[66] A PNA lectin blot confirmed the presence of neoglycoconjugates at high molecular weight (>60 kDa) in lanes where the initiator and glycan were present and at longer irradiation times. The observed molecular weight range for the lectin-reactive bands cannot be accounted for by attachment of the SGP glycan, suggesting that these bands may indicate the presence of α-synuclein oligomers. Although we observe some monomer remained (Figure 4A), lectin staining indicated that the glycosylated forms are predominantly at higher molecular weight consistent with tetramers, or larger, oligomers. Previous work has found that recombinantly expressed monomers of α-synuclein can readily aggregate into amyloid-like fibrils in vitro, and SDS-stable α-synuclein bands >55 kDa have been observed.[67,68]

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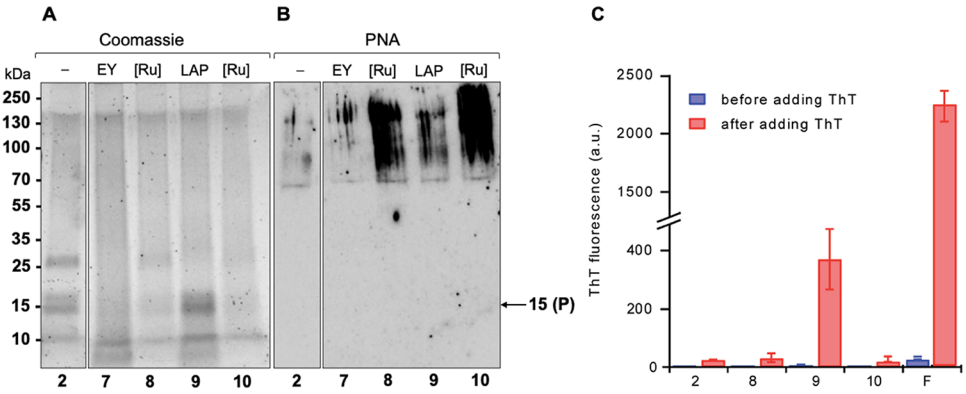


Figure 4. Glycoconjugates of α-synuclein and SGP-glycoside **13**. α-synuclein glycoconjugates were probed with biotinylated PNA lectin to confirm the attachment of the desired glycan. Samples were loaded on (A) SDS-PAGE (12%) under reducing conditions and stained with Coomassie. (B) Lectin blot of the transferred gel was probed with PNA and developed with streptavidin-HRP. Lane 2 represents the unlabeled α-synuclein, and lanes 7, 8, 9, and 10 represent conjugation using EY (20 min), LAP (1 h), and [Ru] (30 min). Lanes 8 and 10 represent the [Ru] initiator used in 0.5 or 2.0 equiv to the thiol concentration, respectively. The expected MW of α-synuclein (15 kDa) is indicated. (C) ThT assay for α-synuclein conjugates along with WT fibrils (F) as a positive control. Entries 8−10 indicate the samples with LAP and [Ru] as initiators corresponding to those in panels A and B. Entry 2 represents the α-synuclein negative control. The full gel image is provided in Figure S8.

We again used RP-HPLC to provide insight into the extent of formation of α-synuclein SGP conjugates (Figure S8). Consistent with lectin staining, we found that conditions with all three initiators showed formation of glycoconjugates. Based on consumption of the starting material, EY and [Ru] initiators showed better conversion (23 and 20%, respectively) along with unreacted starting materials (Table S7). In the case of LAP, we observed starting material consumption and a small product peak (6%) as well as a new, slower migrating peak at 10.9 min (54%), which we interpreted as an oligomer of α-synuclein (Figure S8). In control experiments we observed that oligomers of α-synuclein retained by 100 K MWCO filters showed longer retention times by HPLC (>9.4 min retention time, see Figure S11), similar to the peaks observed after LAP treatment. Together, these data support that TEC conjugation of SGP to α-synuclein was successful and that the conjugates themselves may have increased propensity for oligomer formation, and decreased formation of fibrillar aggregates.

ThT Assay for α-Synuclein Conjugates. Previous investigations of O-GlcNAc modified α-synuclein conjugates have observed that glycoconjugates form high molecular weight oligomers that have reduced propensity to form fibrillar aggregates.[69−73] To investigate the properties of our α-synuclein glycoconjugates, we used a thioflavin T (ThT) assay using wildtype fibrils as positive control (Figure 4C). The ThT chromophore should have increased fluorescence in the presence of fibrillar aggregates. We observed that ThT fluorescence was low for unmodified α-synuclein, and the fibril positive control (F) showed a large increase with ThT addition. The ThT fluorescence assay could not be used for conjugates made with EY, as the initiator gave significant background in the same region (450−485 nm), a potential limitation of this strategy for ThT analysis. Conjugates made with the [Ru] initiator did not show significant increases in ThT signal and remained at the same or lower values compared to unmodified α-synuclein. While conjugates made with the LAP initiator did show higher fluorescence, control experiments with LAP and α-synuclein alone showed a similar increase in fluorescence (Figure S9).[74] Furthermore, since certain small molecules (e.g., dopamine) are known to increase α-synuclein aggregation, we concluded that the increase in ThT signal in these samples was an artifact of LAP interaction with the protein.[75] Taken with our lectin blotting and HPLC data, we conclude that glycoconjugates of synuclein with N-linked glycans had increased propensity for high-molecular weight oligomer formation, but this was not due to fibril formation based on ThT assays. Thus, these neoglycoconjugates, with the exception of LAP-treated samples, showed reduced fibrillar aggregation, consistent with observations with O-GlcNAc modified α-synuclein investigated by Pratt and coworkers.[76,71,72]

CONCLUSIONS

We have developed a facile linker strategy for the preparation of glycoconjugates. The strategy relies on the use of naturally occurring or engineered Cys residues in the target protein but requires minimal manipulation of the glycan. Glycans were used without any protecting group manipulations and could be obtained from biological sources and used as reducing sugars in the strategy. The linkers used here, *O*-allyl-*N*-methylhydroxylamine **1** and allyl alcohol **2** are both commercially available and provide access to either *N*- or *O*-glycosides, and our model compounds demonstrate that they are compatible with typical N-link reducing end residues. The glycosides generated by this method retain the reducing end residue in its closed-ring form. The *O*-glycosides were generated primarily as β-anomers, whereas the *N*-glycosides were composed of a mixture of anomers. We also observed contamination of the final product with the photoinitiators could interfere with some assays, which may be a potential disadvantage of the approach. Coupling efficiency of the TEC reaction ranged from 70 to 90% conversion depending on the target. The advantage of this strategy is that it uses mild reaction conditions on both the glycan and protein target, allowing its application to proteins which may be sensitive to harsh conditions.

EXPERIMENTAL PROCEDURES

General Methods. Reagents and solvents were purchased from commercial sources and used without further purification. The *O*-allyl-*N*-methylhydroxylamine oxalate salt (**2**) was purchased from A1 Biochem Laboratories. L-Cysteine, was purchased from Aldrich. *N*-acetyl-D-glucosamine and *N*,*N*′-

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diacetylchitobiose were purchased from Carbosynth. mPEG-Alkene (5 K) and (10 K) were purchased from Biopharma PEG. Lysozyme (from chicken egg white), Bovine Serum Albumin (lyophilized powder), α-synuclein and bvPrP (N197C) were prepared as described elsewhere.[65] Radical initiators 2-hydroxy-4′-(2-hydroxyethoxy)-2-methylpropiophenone (HHMP), 2,2-dimethoxy-2-phenylacetophenone (DPAP or DMPA), Lithium phenyl-2,4,6-trimethylbenzoylphosphinate (LAP), Tris(2,2′-bipyrazine)ruthenium bis-(hexafluorophosphate) $[Ru(bpz)_3][PF_6]_2$, Eosin Y disodium salt (EY) were all purchased from Sigma-Aldrich. TLC was performed on precoated silica TLC plates (Merck TLC silica gel 60 F254) and visualized by ultraviolet light (254 nm) and by charring with 10% H_2SO_4 in EtOH or 0.2% (w/v) ninhydrin in EtOH, *p*-anisaldehyde stain, or anisidine phthalate in EtOH. Purification was carried out using C18 Sep-pak plus (prepacked catridges, Waters) or C18 nonend-capped 23% silica 40−63 μm (SiliCycle) packed columns. Flash chromatography was performed with silica gel (Silica gel, F60, 40−63 μm particle size, Aldrich). Proton (^{1}H) and Carbon (^{13}C) NMR spectra were recorded on Varian 500 (500 MHz for ^{1}H and 126 MHz for ^{13}C) along with 2D-NMR (COSY, HSQC, HMBC) as required. High-Resolution Mass Spectrometry (HRMS) was performed on an Agilent Technologies TOF 6220 spectrometer. PNGase-F enzyme, β-lactoglobulin (BLG), and recombinant GTB were expressed in *E. coli* and purified by FPLC (AKTAprime plus) using a Ni-NTA column (Gen) and confirmed by SDS-PAGE. Bradford reagent (Sigma-Aldrich) was used for Bradford assay, performed on a 96-well plate (Costar CLS 3799) and read on a fluorescence plate reader (Abs 595 nm). Mini-Protean TGX precast gels (10% and 12%), 10× Tris/Glycine/SDS buffer, BioSafe Coomassie stain, nitrocellulose membranes (0.2 μm), 2× and 4× Laemmli sample buffer, and Clarity Western ECL substrate were purchased from Bio-Rad. Thermo Scientific PageRuler Plus Prestained Protein Ladder (Cat No. 26619) was used for SDS-PAGE. Biotinylated Elderberry bark lectin (SNA) and biotinylated Peanut Agglutinin lectin (PNA) were purchased from Vector Laboratories. Streptavidin-HRP (DY998) was purchased from R&D systems. Gels and blots were visualized using ChemiDoc MP imaging system (Bio-Rad). The following light sources were used for the photochemical reactions: UV flood lamp source (385−400 nm), Xenon arc lamp (Sutter Instrument Company, 100−240 V, 50/60 Hz) with filter #3 for Blue LED (430−510 nm), and filter #5 for Green LED (485−585 nm).

General Procedure for *N*-Glycosidation. *O-Allyl-N-methyl-N-(2-aminoacetyl-2-deoxy-D-glucopyranosyl)-hydroxylamine (**4a−g**). Method a.* To a solution of *N*-acetyl-D-glucosamine (15 mg, 0.07 mmol) in 1:1 (v/v) DMF/AcOH (1 mL), *O*-allyl-*N*-methylhydroxylamine oxalate salt (18 mg, 0.1 mmol) was added and stirred at room temperature for 20 h. The reaction progress was monitored by TLC (2:1, MeOH:EtOAc). The reaction mixture was quenched with MeOH and concentrated. The resulting residue was purified by C18 silica column chromatography (2:1, H_2O:MeOH) to give compound **4** (30 mg, 91%; α:β 76:24); ^{1}H NMR (500 MHz, CD_3OD) δ 5.93 (ddt, 1H, J = 17.5, 10.5, 6.5 Hz, H-c), 5.49−5.24 (dd, 1H, $J_{a,c}$ = 17.5, $J_{a,b}$ = 1.5 Hz, H-a), 5.42−5.15 (dd, 1H, $J_{b,c}$ = 10.5, $J_{b,a}$ = 1.5 Hz, H-b), 5.09 (d, 1H, J = 3.5 Hz, H-1α), 4.58 (d, 1H, J = 8.5 Hz, H-1β), 4.56−4.13 (m, 2H, H-d, e), 3.87−3.77 (m, 4H, H-5, 2, 6, 6), 3.71−3.68 (m, 1H, H-3), 3.37−3.36 (m, 1H, H-4), 2.71 (s, 3H, CH_3), 1.98 (s, 3H, $COCH_3$); ^{13}C NMR (126 MHz, CD_3OD) δ 173.6 (C=O), 135.3 (CH-c), 118.9 (CH_2-a, b), 97.1 (C1β), 92.6 (C1α), 75.2 (CH_2-d, e), 73.1 (C2), 72.6 (C3), 72.5 (C4), 62.8 (C6), 55.9 (C5), 38.8 (N-CH_3), 22.8 (CH_3); HRMS (ESI) m/z calcd for $C_{12}H_{23}N_2O_6$ $[M + H]^+$ 291.1551, found 291.1551.

Method b. The *O*-allyl-*N*-methylhydroxylamine oxalate salt (27 mg, 0.15 mmol) and *N*-acetyl-D-glucosamine (37 mg, 0.16 mmol) were dissolved in 5:1 (v/v) MeOH/AcOH (5 mL) and the resulting mixture was stirred at room temperature for 72 h. The reaction progress was monitored by TLC (2:1, MeOH:EtOAc). The reaction mixture was quenched with MeOH and concentrated. The resulting residue was purified by C18 silica column chromatography (2:1, MeOH:EtOAc) to give compound **4** (30 mg, 75%; α:β 71:29).

Method c. The *O*-allyl-*N*-methylhydroxylamine oxalate salt (26 mg, 0.15 mmol) and *N*-acetyl-D-glucosamine (30 mg, 0.14 mmol) were dissolved in 2 M aq. NH_4OAc/AcOH pH 4.5 buffer (1.4 mL). The reaction mixture was stirred at 40 °C for 72 h as the reaction progress was monitored by TLC (2:1, MeOH:EtOAc). The reaction mixture was quenched with MeOH and concentrated. The resulting residue was purified by C18 silica column chromatography (2:1, MeOH:H_2O) to give compound **4** (30 mg, 76%; α:β 59:41).

Method d. To a solution of *N*-acetyl-D-glucosamine (30 mg, 0.14 mmol) in H_2O (0.5 mL), *O*-allyl-*N*-methylhydroxylamine oxalate salt (62.5 mg, 0.35 mmol) in 0.1 M NaOAc/AcOH pH 4.0 buffer (0.5 mL) was added at 0 °C. The reaction mixture was stirred at room temperature for 20 h. The reaction progress was monitored by TLC (6:3:3:2 EtOAc/AcOH/MeOH/H_2O) and the mixture was concentrated. The resulting residue was purified by C18 silica column chromatography (2:1, H_2O:MeOH) to give compound **4** (29 mg, 73%; α:β 66:34).

Method e. To a solution of *O*-allyl-*N*-methylhydroxylamine oxalate salt (28 mg, 0.15 mmol) in acetonitrile (5 mL), *N*-acetyl-D-glucosamine (38.5 mg, 0.17 mmol) was added. H_2O (2.5 mL) was added to yield a clear solution, followed by addition of glacial AcOH (75 μL). The reaction mixture was stirred at room temperature for 16 h as the reaction progress was monitored by TLC (5:1, DCM:MeOH). The reaction mixture was quenched with MeOH and concentrated. The resulting residue was purified by C18 silica column chromatography (5:1, DCM:MeOH) to give compound **4** (40 mg, 87%; α:β 58:42).

Method f. The *O*-allyl-*N*-methylhydroxylamine oxalate salt (96 mg, 0.54 mmol) and *N*-acetyl-D-glucosamine (12 mg, 0.05 mmol) were dissolved in 3:1 (v/v) DMF/AcOH (0.5 mL). The reaction mixture was stirred at 40 °C for 72 h as the reaction progress was monitored by TLC (6:3:3:2 EtOAc/AcOH/MeOH/H_2O). The reaction mixture was quenched with MeOH and concentrated. The resulting residue was purified by C18 silica column chromatography (2:1, H_2O:MeOH) to give compound **4** (13 mg, 83%; α:β 59:41).

Method g. The *O*-allyl-*N*-methylhydroxylamine oxalate salt (64 mg, 0.36 mmol) and *N*-acetyl-D-glucosamine (8 mg, 0.03 mmol) were dissolved in 1:1 (v/v) 1,4-dioxane/NH_4OAc pH 4.5 buffer (1.1 mL). The reaction mixture was stirred at 40 °C for 72 h as the reaction progress was monitored by TLC (6:3:3:2 EtOAc/AcOH/MeOH/H_2O). The crude mixture was quenched with MeOH and concentrated. The resulting residue was purified by C18 silica column chromatography (2:1, H_2O:MeOH) to give compound **4** (10 mg, 95%; α:β 51:49).

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O-Allyl-N-methyl-N-[2-Acetamido-4-O-(2-acetamido-2-deoxy-β-D-glucopyranosyl)-2-deoxy-D-glucopyranosyl]-hydroxylamine (8). Method a. The *O*-allyl-*N*-methylhydroxylamine oxalate salt (5 mg, 0.02 mmol) and *N,N'*-diacetylchitobiose (13.2 mg, 0.03 mmol) were dissolved in 5:1 (v/v) MeOH/AcOH (5 mL) and the resulting mixture was stirred at room temperature for 90 h. The reaction progress was monitored by TLC (2:1, MeOH:EtOAc). The reaction mixture was quenched with MeOH and concentrated. The resulting residue was purified by C18 silica column chromatography (2:1, MeOH:EtOAc) to give compound **8** (11 mg, 79%; α:β 77:23); ^{1}H NMR (500 MHz, D_2O) δ 5.99 (ddt, 1H, *J* = 17.0, 10.5, 6.5 Hz, H-c), 5.41 (d, 1H, $J_{a,c}$ = 17.0 Hz, H-a), 5.35 (d, 1H, $J_{a,b}$ = 10.5 Hz, H-b), 5.26 (d, 0.75H, *J* = 2.0 Hz, H-1α), 4.76 (d, 0.25H, *J* = 7.5 Hz, H-1β), 4.65 (d, 1H, *J* = 8.5 Hz, H-1'β), 4.28 (d, 1H, *J* = 10.0 Hz, H-4), 4.24 (d, 1H, *J* = 6.5 Hz, H-d), 4.19 (d, 1H, *J* = 6.5 Hz, H-e), 4.08 (t, 1H, *J* = 10.0 Hz, H-5), 3.99 (d, 1H, *J* = 12.0 Hz, H-3'), 3.93 (d, 1H, *J* = 12.0 Hz, H-5'), 3.84−3.79 (m, 1H, H-2'), 3.75−3.71 (m, 2H, H-6), 3.69−3.62 (m, 3H, H-2, 6'), 3.59−3.50 (m, 2H, H-3, 4'), 2.81 (s, 3H, CH_3), 2.14 (s, 3H, $COCH_3$), 2.10 (s, 3H, $COCH_3$); ^{13}C NMR (126 MHz, D_2O) δ 173.6 (C=O), 173.4 (C=O), 133.6 (CH-c), 121.0 (CH_2-a, b), 102.4 (C1β'), 101.9 (C1β), 96.3 (C1α), 92.5 (C4), 80.1 (C2), 76.9 (C4'), 74.9 (CH_2-d, e), 74.7 (C5'), 74.4 (C6'), 70.6 (C3), 61.5 (C3'), 61.2 (C6), 56.6 (C2'), 52.7 (C5), 38.4 (N-CH_3), 23.3 (CH_3), 23.1 (CH_3); HRMS (ESI) *m/z* calcd for $C_{20}H_{36}N_3O_{11}$ [M + H]$^+$ 494. 2344, found 494.2353; calcd for $C_{20}H_{35}N_3O_{11}Na$ [M + Na]$^+$ 516.2164, found 516.2161.

Method b. The *O*-allyl-*N*-methylhydroxylamine oxalate salt (2.7 mg, 0.01 mmol) and *N,N'*-diacetylchitobiose (7.1 mg, 0.02 mmol) were dissolved in 9:1 (v/v) MeOH/$CHCl_3$ (0.5 mL). AcOH (1 equiv) was added and the reaction mixture was stirred at room temperature for 72 h. The reaction progress was monitored by TLC (2:1, MeOH:EtOAc). The reaction mixture was quenched with MeOH and concentrated. The resulting residue was purified by C18 silica column chromatography (2:1, MeOH:EtOAc) to give compound **8** (6 mg, 81%; α:β 76:24).

Method c. The *O*-allyl-*N*-methylhydroxylamine oxalate salt (11.5 mg, 0.06 mmol) and *N,N'*-diacetylchitobiose (25 mg, 0.05 mmol) were dissolved in 2 M aq. NH_4OAc/AcOH pH 4.5 buffer (0.6 mL). The reaction mixture was stirred at 40 °C for 72 h as the reaction progress was monitored by TLC (6:3:3:2 EtOAc/AcOH/MeOH/H_2O). The reaction mixture was quenched with MeOH and concentrated. The resulting residue was purified by C18 silica column chromatography (2:1, H_2O:MeOH) to give compound **8** (10 mg, 35%; α:β 59:41).

Procedure for *O*-Glycosidation. *N-(2-Acetamido-2-deoxy-β-D-glucopyranosyl-p-toluenesulfono-hydrazide (5).* *N*-Acetyl-D-glucosamine (5 g, 22.6 mmol) and *p*-toluenesulfonylhydrazide (4.64 g, 24.9 mmol) were suspended in a mixture of DMF (10 mL) and distilled water (5 mL) in a 50 mL polypropylene tube. Glacial acetic acid (1.0 mL, 18 mmol) was then added and the mixture was incubated at 37 °C (without stirring) until all solid was dissolved for 2 days. The clear homogeneous solution was then poured into diethyl ether (800 mL) and stirred vigorously for 20 h. The precipitate was collected, dissolved in distilled water (20 mL) and subsequently freeze-dried, affording the desired compound **5** as a white solid (8 g, 91%, α:β = 1:11); ^{1}H NMR (500 MHz, CD_3OD) δ 7.71 (d, 2H, *J* = 8.5 Hz, Ar−H), 7.36 (d, 2H, *J* = 8.0 Hz, Ar−H), 3.92 (d, 1H, *J* = 9.5 Hz, H-1β), 3.87 (dd, 1H, *J* = 11.0 Hz, H-6), 3.60−3.57 (m, 1H, H-6), 3.45 (t, 1H, *J* = 9.5 Hz, H-2), 3.40−3.36 (m, 1H, H-3) 3.19−3.17 (m, 2H, H-4, 5), 2.42 (s, 3H, Ar−CH_3), 1.99 (s, 3H, $COCH_3$); ^{13}C NMR (126 MHz, CD_3OD) δ 173.9 (C = O), 145.1 (Ar-C-S), 137.2 (Ar-C-CH_3), 130.7 (Ar−CH), 130.5 (Ar−CH), 129.1 (Ar−CH), 129.0 (Ar−CH), 91.9 (C1β), 78.9 (C5), 76.3 (C3), 72.4 (C4), 63.2 (C6), 55.0 (C2), 23.0 ($COCH_3$), 21.5 (Ar−CH_3); HRMS (ESI) *m/z* calcd for $C_{15}H_{24}N_3O_7S$ [M + H]$^+$ 390.1329, found 390.1327; calcd for $C_{15}H_{23}N_3O_7SNa$ [M + Na]$^+$ 412.1149, found 412.1147.

O-Allyl 2-Acetamido-2-deoxy-β-D-glucopyranoside (6). To a solution of compound **5** (50 mg, 0.13 mmol) in anhydrous DMF (1.0 mL), predried allyl alcohol (5.1 mmol) was added, followed by the addition of *N*-iodosuccinimide (57 mg, 0.26 mmol) at room temperature. After 10 min of stirring, the reaction mixture was quenched with Amberlite resin (OH^-) and the solution was stirred until the yellow color disappeared. The resin was then filtered and the solvent was removed under reduced pressure. The resulting residue was purified by flash column chromatography (8% MeOH/DCM) to afford compound **6** as a white solid (126.8 mg, 38%, α:β = 1:6); ^{1}H NMR (500 MHz, D_2O) δ 5.98 (ddt, 1H, *J* = 17.0, 10.5, 6.0 Hz, H-c), 5.38 (d, 1H, $J_{a,c}$ = 17.5 Hz, H-a), 5.33 (d, 1H, $J_{b,c}$ = 10.5 Hz, H-b), 4.63 (d, 1H, *J* = 8.5 Hz, H-1β), 4.41 (dd, 1H, $J_{d,e}$ = 13.0 Hz, $J_{d,c}$ = 5.0 Hz, H-d), 4.23 (dd, 1H, $J_{e,d}$ = 13.0 Hz, $J_{e,c}$ = 6.5 Hz, H-e), 4.00−3.98 (m, 1H, H-6), 3.96−3.91 (m, 1H, H-6), 3.84−3.81 (m, 1H, H-5), 3.80−3.76 (m, 1H, H-2), 3.63−3.59 (m, 1H, H-3), 3.53−3.51 (m, 1H, H-4), 2.11 (s, 3H, $COCH_3$); ^{13}C NMR (126 MHz, D_2O) δ 175.6 (C=O), 134.3 (CH-c), 119.1 (CH_2-a, b), 101.1 (C1β), 76.9 (C3), 74.9 (C2), 71.4 (CH_2-d, e), 70.9 (C4), 61.7 (C6), 56.5 (C5), 23.1 (CH_3); HRMS (ESI) *m/z* calcd for $C_{11}H_{19}NO_6Na$ [M + Na]$^+$ 284.1105, found 284.1103.

Procedure for Thiol−Ene Coupling with L-Cysteine. *S-[3-(Methylamino)oxy]-L-cysteine (10).* The *O*-allyl-*N*-methylhydroxylamine oxalate salt (5 mg, 0.03 mmol) and L-cysteine hydrochloride (34.2 mg, 0.3 mmol) were dissolved in anhydrous MeOH (0.5 mL) in a quartz tube. The solution was degassed and filled with argon. The reaction mixture was irradiated in a UV photoreactor (310 nm) at room temperature for 2 h as the progress of reaction was monitored by TLC (6:3:3:2 EtOAc/AcOH/MeOH/H_2O). The mixture was concentrated and the resulting residue was purified by C18 silica column chromatography (2:1, MeOH:H_2O) to give compound **10** (8 mg, 65% crude yield; with 22% cystine due to oxidation based on NMR); ^{1}H NMR (500 MHz, D_2O) δ 3.99 (dd, *J* = 7.0, 4.0 Hz, Cys-H_α), 3.96 (t, 2H, $J_{b,c}$ = 6.0 Hz, H-c), 3.19 (d, 1H, *J* = 6.5 Hz, Cys-H_β), 3.13 (d, 1H, *J* = 7.0 Hz, Cys-H_β), 2.78 (s, 3H, N-CH_3), 2.75 (t, 2H, $J_{a,b}$ = 7.0 Hz, H-a), 1.97 (tt, 2H, $J_{a,b}$ = 7.0 Hz, $J_{b,c}$ = 6.0 Hz, H-b); ^{13}C NMR (126 MHz, D_2O) δ 173.8 (C=O), 72.3 (CH_2-c), 54.5 (Cys-C_α), 37.9 (CH_3), 33.0 (Cys-C_β), 28.8 (CH_2-a), 28.5 (CH_2-b); HRMS (ESI) *m/z* calcd for $C_7H_{17}N_2O_3S$ [M + H]$^+$ 209.0954, found 209.0952; calcd for $C_7H_{16}N_2O_3SNa$ [M + Na]$^+$ 231.0774, found 231.0777.

S-[3-Hydroxypropyl]-L-cysteine (11). Allyl alcohol (5 mg, 0.09 mmol) and L-cysteine hydrochloride (31.3 mg, 0.3 mmol) were dissolved in anhydrous MeOH (0.5 mL) in a quartz tube. The solution was degassed and filled with argon. The reaction mixture was irradiated in a UV photoreactor (310 nm) at room temperature for 2 h as the progress of reaction was monitored by TLC (6:3:3:2 EtOAc/AcOH/MeOH/H_2O). The mixture was concentrated and the resulting residue was purified by C18

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silica column chromatography (2:1, MeOH:H_2O) to give compound **11** (28 mg, 76% crude yield; with 19% cystine due to oxidation based on NMR); ^{1}H NMR (500 MHz, D_2O) δ 4.06 (dd, 1H, J = 5.5, 4.5 Hz, Cys-H_α), 3.75 (t, 2H, J = 7.0 Hz, H-c), 3.18 (dd, 1H, J = 15.0, 5.5 Hz, Cys-H_β), 3.09 (dd, 1H, J = 15.0, 4.5 Hz, Cys-H_β), 2.75 (t, 2H, J = 7.0 Hz, H-a), 1.91 (quint, 2H, J = 7.0 Hz, H-b); ^{13}C NMR (126 MHz, D_2O) δ 173.7 (C=O), 60.9 (CH_2-c), 56.6 (Cys-C_α), 32.9 (Cys-C_β), 31.8 (CH_2-b), 28.7 (CH_2-a); HRMS (ESI) m/z calcd for $C_6H_{14}NO_3S$ $[M + H]^+$ 180.0689, found 180.0687; calcd for $C_6H_{13}NO_3SNa$ $[M + Na]^+$ 202.0508, found 202.0505.

Isolation of Complex Glycans Using PNGase-F. PNGase-F was prepared using the expression system, purified by Ni-NTA agarose column, and confirmed by SDS-PAGE at 36 kDa. Deglycosylation of glycoproteins using PNGase-F was done following the NEB protocol[77] under denaturing reaction conditions. Deglycosylation efficiency of PNGase-F was tested on AGP, Fetuin, and sialoglycoprotein (SGP) and analyzed by SDS-PAGE gel. SGP N-glycan with a reducing end was released from sialoglycoprotein[43] to be used for further steps.

Conjugation of *O*-Allyl Linker on SGP Glycan. The best optimized conditions for linker conjugation (**4a**−**g**) were selected for attaching *O*-allyl-*N*-methylhydroxylamine linker (**1**) to the reducing end of SGP glycan. The complex glycan with the terminal double bond can then be used for TEC with a cysteine thiol on proteins.

General Procedure for Thiol−Ene Coupling between Protein and mPEG-Alkene (5 or 10 K). *I—Lithium Phenyl-2,4,6-trimethylbenzoylphosphinate (LAP) as the Initiator.* A protein sample (2 mg, 80 μM) and mPEG-Alkene (5 or 10 K) (4.2 mg, 140 μM) were dissolved in 500 μL of buffer (250 mM NaOAc pH 5.0 or 250 mM Tris pH 6.8) or MQ H_2O in a vial with a flea bar and screw cap. From a 5 mM LAP stock solution (3 mg dissolved in 2 mL MQ H_2O, stored in an Amber Eppendorf), 300 μL was added to the reaction mixture to obtain a final conc of 1.5 mM LAP and the mixture was made up to 1 mL final reaction volume. Similarly, 400 μL was added to the reaction mixture to obtain a final conc of 2.0 mM LAP and the mixture was made up to 1 mL final reaction volume. The reaction mixture was mixed well. The stir plate with the reaction vial was set up at a distance of 2 cm from the UV 385−400 nm source. The mixture was stirred with irradiation over a period of 30 min to 1 h.

II—Tris(2,2′-bipyrazine) Ruthenium Bis-(hexafluorophosphate) or [Ru(bpz)$_3$][PF6]$_2$ as the Initiator. A protein sample (2 mg, 80 μM) and mPEG-Alkene (5 or 10 K) (3.5 mg, 230 μM) were dissolved in 200 μL of MQ H_2O in a vial with a flea bar and screw cap. From 1.2 mM [Ru] stock solution (1 mg dissolved in 1 mL MQ H_2O, stored in Amber Eppendorf), 25 μL was added to the reaction mixture to make a final conc of 0.03 mM and the mixture is made up to 1 mL final reaction volume. The reaction mixture was mixed well. The stir plate with the reaction vial was set up at a distance of 2 cm from the Blue LED source (470 nm), filter #3 on Xenon arc lamp. The mixture was stirred with irradiation over a period of 10 min to 1 h.

III—Eosin-Y (EY) as the Initiator. A protein sample (1 mg, 80 μM) and of mPEG-Alkene (5 or 10 K) (2.8 mg, 300 μM) were dissolved in 200 mL of MQ H_2O in a vial with a flea bar and screw cap. From 2.9 mM EY stock solution (4 mg dissolved in 2 mL MQ H_2O, stored in an amber Eppendorf tube), 35 μL was added to the reaction mixture to make a final conc of 0.1 mM EY and the mixture is made up to 1 mL final reaction volume. Similarly, 69 μL was added to the reaction mixture to make a final conc of 0.2 mM EY and the mixture is made up to 1 mL final reaction volume. The reaction mixture was mixed well. The stir plate with the reaction vial was set up at a distance of 2 cm from the Green LED source (520 nm), filter #5 on Xenon arc lamp. The mixture was stirred with irradiation over a period of 10 min to 1 h.

In I, II and III, controls without the ene linker, initiator or UV were run alongside for the same duration. A 20−50 μL aliquot was taken after each interval and used to run SDS-PAGE. Gels were analyzed for the appearance of new bands indicating conjugation.

HPLC Analysis for BSA-mPEG Conjugates. The glycosidation products **4a**−**g** and BSA-mPEG conjugates (Figure 2) were characterized by HPLC on a Waters HPLC equipped with a DAD detector. The separation was achieved on a Waters Xterra RP18 column (3.5 μm, 4.6 × 150 mm) with the following eluents: A: H_2O/MeCN (9/1), 0.1% TFA and B: MeCN/H_2O (9/1), 0.1% TFA, and a linear gradient program: 5−100% of B in 15 min, 100−95% B for 5 min, at a flow rate of 1 mL/min, at 25 °C and the injected sample volume was 20 μL.

General Procedure for Thiol−Ene Coupling between Protein and SGP-Glycoside 13. A protein sample (1 mg, 20 μM) was dissolved in 200 μL of MQ H_2O in a vial with a flea bar and screw cap. The SGP-glycoside (9.2 mg) was added to vials with protein for a final reaction volume of 0.5 mL (at 0.2 mM or 0.5 mM final concentration of SGP-glycoside).

I—Lithium Phenyl-2,4,6-trimethylbenzoylphosphinate (LAP) as the Initiator. From a 5 mM LAP stock solution (3 mg dissolved in 2 mL MQ H_2O), a mixture with 1.5 or 2.0 mM final concentration was prepared in 0.5 mL volume. The reaction mixture was then mixed well. The stir plate with the reaction vial was set up at a distance of 2 cm from the UV 385−400 nm flood lamp source and stirred with irradiation over a period of 1−2 h.

II—Tris(2,2′-bipyrazine) Ruthenium Bis-(hexafluorophosphate) or [Ru(bpz)$_3$][PF6]$_2$ as the Initiator. From a 1.2 mM $[Ru(bpz)_3][PF_6]_2$ stock solution (1 mg dissolved in 1 mL MQ H_2O), a mixture with 0.5 or 2.0 equiv. to thiol final concentration was prepared in 0.5 mL volume. The reaction mixture was then mixed well. The stir plate with the reaction vial was set up at a distance of 2 cm from the Blue LED source (470 nm), filter #3 on Xenon arc lamp, and stirred with irradiation over a period of 20 min to 1 h.

III—Eosin-Y (EY) as the Initiator. From a 2.9 mM EY stock solution (4 mg dissolved in 2 mL MQ H_2O), a mixture with 0.1 or 0.2 mM final concentration was prepared in 0.5 mL volume. The reaction mixture was then mixed well. The stir plate with the reaction vial was set up at a distance of 2 cm from the Green LED source (520 nm), filter #5 on Xenon arc lamp, and stirred with irradiation over a period of 20 min to 1 h.

In I, II and III, controls without ene linker, initiator or UV were run alongside for the same duration. A 50 μL aliquot was taken after each interval and used to run SDS-PAGE. Gels were then transferred to nitrocellulose membrane (Lectin blot) and analyzed for the appearance of new bands indicating conjugation using suitable lectins like biotinylated ECA or PNA and streptavidin-HRP.

SDS-PAGE and Lectin Blot for Conjugates. Samples were analyzed by SDS-PAGE using Mini-PROTEAN TGX precast gels 10 and 12%, 10-well comb, 50 μL (Bio-Rad), 10×

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Tris/Glycine/SDS running buffer (Bio-Rad) and 2× or 4× Laemmli sample buffer (Bio-Rad) with DTT. Electrophoresis was performed at 110 mV for 70 min. Gels were stained with BioSafe Coomassie stain (Bio-Rad), followed by destaining in MQ water overnight for comparative analysis of the bands.

For lectin blots, completed gels were transferred to nitrocellulose membrane, 0.2 μm (Bio-Rad) at 25 mV for 16h. 5% BSA was used as blocking solution. Biotinylated Elderberry bark lectin *Sambucus nigra* (SNA) and biotinylated Peanut agglutinin lectin (PNA) were obtained from Vector laboratories. Biotinylated *Erythrina cristagalli*agglutinin (ECA) was a kind gift from Prof. Lara Mahal (University of Alberta). Streptavidin-HRP (R&D systems) were used for developing the membranes. The membranes were then visualized in ChemiDoc MP imaging system (Bio-Rad) using Clarity Western ECL substrate (Bio-Rad). α-synuclein samples were subjected to centrifugation, prior to SDS-PAGE to separate fibrils from monomer and oligomers. No pellets were observed in any of the cases, and the supernatants (soluble fractions) were used for analysis by SDS-PAGE and lectin blots.

MALDI Analysis for BSA Neoglycoconjugates. To prepare the MALDI samples, a two-layer sample preparation method was used. Layer 1 was 10 mg/mL sinapinic acid dissolved in 80% acetone 20% methanol and 1 μL was spotted on target and allowed to dry. Layer 2 solution was 10 mg/mL sinapinic acid dissolved in 50% acetonitrile 50% water with 0.1% TFA. 5 μL of layer 2 solution was mixed with 4 μL of 0.1% TFA (aq) and 1 μL of sample solution; from which 0.8 μL was spotted on top of layer 1 and allowed to dry. This was then rinsed by spotting 2 μL of 0.1% TFA (aq) on top of the dried sample spot. It is kept undisturbed for 10 s, then removed by a gentle stream of air and removed the water droplet with the edge of a lint-free tissue.

HPLC Analysis for BSA-SGP Conjugates. The BSA-SGP conjugates (Figure S6 and Table S5) were characterized by HPLC on a Waters HPLC equipped with a DAD detector. The separation was achieved on a Waters Xterra RP18 column (3.5 μm, 4.6 × 150 mm) with the following eluents: (A) H_2O/MeCN (9/1), 0.1% TFA and (B) MeCN/H_2O (9/1), 0.1% TFA, and a linear gradient program: 5−100% of B in 15 min, 100−95% B for 5 min, at a flow rate of 1 mL/min, at 25 °C and the injected sample volume was 20 μL.

Production of Recombinant α-Synuclein with a Cysteine (α-synC). Briefly, the open reading frame encoding recombinant α-synuclein with a cysteine residue (α-synC) appended to the native C-terminus was inserted into a pET21b expression vector and used to transform *E. coli* Rosetta 2 (DE3) cells to ampicillin resistance. Protein expression was induced at the mid logarithmic growth stage of the culture with a final concentration of 1 mM isopropyl β-D-1-thiogalactopyranoside (IPTG), and cultures were harvested 6 h later. Bacterial pellets were resuspended in an osmotic-shock buffer (30 mM Tris pH 7.2, 40% (w/v) sucrose, 2 mM EDTA). At this point, and in all subsequent buffers, 2 mM Tris(2-carboxyethyl)phosphine (TCEP) was added when preparing the protein to ensure the suflhydryl group of the cysteine side-chain remained reduced. Cells were pelleted by centrifugation (21,036*g*, 20 min) and then resuspended in 90 mL ice cold Milli-Q H_2O containing 37.5 μL of saturated $MgCl_2$. The sample was kept on ice for 10 min, then centrifuged again (21,036*g*, 20 min) to pellet the cells. The resulting supernatant containing proteins released from the *E. coli* periplasm was subjected to ammonium sulfate precipitation: saturated ammonium sulfate was added and brought to a final concentration of 40% w/v to precipitate the protein and incubated overnight at 4 °C with gentle stirring. The precipitate was collected by centrifugation (75,600*g*, 60 min) then subjected to further purification by anion exchange chromatography.

The 40% ammonium sulfate pellet of α-synC was dissolved it in ice-cold Q-Sepharose buffer A (20 mM Tris pH 8.0) then applied to a Q-Sepharose FF (GE Life Sciences) column pre-equilibrated with buffer A using an ÄKTA Purifier 100 FPLC system. Unbound proteins were washed off with 10 column-volumes of buffer A, then α-synuclein was eluted at approximately 300 uM salt using a gradient of 0−50% of buffer B (20 mM Tris, 1 M NaCl, pH 8.0). Peak fractions were analyzed by SDS-PAGE; those containing α-synuclein were pooled and dialyzed against Milli-Q water to remove salts. Aliquots of the dialyzed sample containing approximately 5 mg of α-synC were frozen in liquid nitrogen and freeze-dried. All samples were quantified spectrophotometrically using Beer's law, measuring the absorbance at 280 nm with a UV/vis spectrophotometer (Cary 60, Agilent Technologies) and using a theoretical extinction coefficient of 5960 M^{-1} cm^{-1}.

HPLC Analysis for α-*syn*-SGP Conjugates. The α-*syn*-SGP conjugates (Figure S8 and Table S7) were characterized by HPLC on a Waters HPLC equipped with a DAD detector. The separation was achieved on a Waters Xterra RP18 column (3.5 μm, 4.6 × 150 mm) with the following eluents: (A) H_2O/MeCN (9/1), 0.1% TFA and (B) MeCN/H_2O (9/1), 0.1% TFA, and a linear gradient program: 5−100% of B in 15 min, 100−95% B for 5 min, at a flow rate of 1 mL/min, at 25 °C and the injected sample volume was 20 μL.

ThT Assay for α-Synuclein Conjugates. ThT assays of α-synuclein fibrillization were performed using a Synergy H1 microplate reader (BioTek). For each experimental condition, 100 μL of 30 μM for each sample in PBS containing 20 μM ThT (Sigma-Aldrich) was placed in the wells of a black, sealed, clear-bottomed 96-well plate (Greiner Bio-One). 20 μM α-synuclein fibrils was measured as a control. Microplates were incubated at 37 °C with continuous shaking, and ThT fluorescence was quantified at by exciting at 440 nm and measuring emission at 490 nm. A total of 19 replicates were measured for each sample, and the results were averaged, error was calculated as SEM.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.bioconjchem.5c00138.

Supplementary data including figures and tables, NMR spectra, mass spectra, and HPLC chromatograms (PDF)

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Notes

The authors declare no competing financial interest.

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